Emotion Aware Face Analysis System

Chiranshu Kataria* Daxal Prajapati* Himanshu* Naveen Kumar* Ujjawal*
Indian Institute of Technology Kanpur
{chiranshu22,daxal22,himanshu22,naveenk22,ujjawal22}@iitk.ac.in

GitHub Repository: https://github.com/DaxalP/EE655-Group13

Dataset Link: https://www.kaggle.com/datasets/ananthu017/emotion-detection-fer?select=test

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Abstract

Facial emotion recognition (FER) from static images is a fundamental problem in affective computing with applications spanning healthcare, education, and human-computer interaction. While transfer learning from large-scale RGB datasets such as ImageNet has become the dominant paradigm, we demonstrate that this approach is sub-optimal when applied to inherently grayscale facial expression datasets. In this work, we propose **GrayFER**, a domain-aligned CNN architecture specifically designed for grayscale FER. Our approach introduces three key contributions: (1) a hierarchical dual-convolution block design with progressive filter expansion (32→64→128) that captures multi-scale facial features without channel redundancy, (2) an adaptive class-balancing augmentation pipeline that addresses the inherent imbalance in emotion datasets through targeted synthetic generation for underrepresented classes, and (3) a systematic empirical analysis comparing eight model configurations across custom CNNs and transfer learning baselines (VGG16, ResNet101, EfficientNetV2B2). GrayFER achieves **82.03%** test accuracy on a four-class emotion classification task (happy, sad, neutral, surprise), outperforming all transfer learning baselines by over 30 percentage points and simpler CNN architectures by 10–16 points, while maintaining an efficient inference time of **4.2 ms** per image on a single GPU. Our results provide strong evidence that modality-aligned, task-specific architectures outperform generic pretrained models when the target domain diverges significantly from the pretraining distribution.

1 Introduction

Facial Expression Recognition (FER) is a cornerstone task in affective computing, aiming to classify human emotions from facial images into discrete categories. The ability to automatically detect emotions has transformative implications: in healthcare for monitoring mental health conditions [1], in education for gauging student engagement [15], and in automotive systems for driver fatigue detection [9]. The global affective computing market, valued at $62.53 billion in 2023, is projected to reach $388.28 billion by 2030, reflecting the growing industrial demand for robust emotion-aware systems [6].

Despite remarkable progress in deep learning for visual recognition, FER remains challenging due to several factors: (a) high inter-class similarity between emotions such as *sad* and *neutral*, (b) significant intra-class variation caused by differences in age, ethnicity, and lighting, (c) inherent label ambiguity since facial expressions are not universally indicative of specific emotions [1], and (d) class imbalance in publicly available datasets.

A dominant approach in recent years has been to leverage transfer learning from models pretrained on ImageNet [4], such as VGG [19], ResNet [7], and EfficientNet [21]. While these models learn rich hierarchical feature representations, they are pretrained on RGB natural images—a fundamentally different distribution from the grayscale, tightly-cropped facial images typical of FER datasets. This distributional mismatch, which we term the *modality gap*, can severely degrade transfer learning performance, as we empirically demonstrate in this work.

Key Insight. We hypothesize that for grayscale FER tasks, a lightweight domain-aligned CNN architecture that directly learns from single-channel inputs will outperform heavyweight transfer learning models that carry unnecessary RGB-specific feature extractors. Our experiments validate this hypothesis, showing a performance gap exceeding 30 percentage points between our custom grayscale CNN and frozen ImageNet-pretrained baselines.

Contributions. Our main contributions are:

1. **GrayFER Architecture:** A purpose-built grayscale CNN with dual-convolution blocks, progressive channel expansion, strategic dropout scheduling (p=0.25 in convolutional layers, p=0.5 in dense layers), and batch normalization—achieving 82.03% test accuracy.
2. **Adaptive Augmentation Pipeline:** A class-aware augmentation strategy that selectively generates synthetic samples for underrepresented emotions, improving class balance without manual data collection.
3. **Comprehensive Benchmark:** A systematic comparison of eight model configurations spanning three families (simple CNNs, medium CNNs, and ImageNet-pretrained models) across both RGB and grayscale input modes, providing actionable insights on when transfer learning helps versus hurts.
4. **Modality Gap Analysis:** An empirical characterization of the performance degradation caused by applying RGB-pretrained models to grayscale data, with analysis of per-class failure modes.

2 Related Work

2.1 Facial Expression Recognition

Early FER systems relied on handcrafted features such as Local Binary Patterns (LBP) [17], Histogram of Oriented Gradients (HOG) [3], and Gabor filters [13] combined with classifiers like SVMs. While interpretable, these methods struggled with illumination and pose variations. The advent of deep learning brought significant improvements: DeepFace [20] and FaceNet [16] demonstrated that CNNs could learn discriminative facial features end-to-end. For emotion classification specifically, the FER-2013 challenge [5] established a benchmark that spurred development of increasingly complex architectures.

Recent works have explored attention mechanisms [22] to focus on salient facial regions, multi-task learning that jointly predicts emotion and action units [12], and Graph Neural Networks that model geometric relationships between facial landmarks [23]. Our work differs in its focus on the underexplored modality alignment problem—demonstrating that architecture design aligned with data characteristics outperforms indiscriminate application of state-of-the-art models.

2.2 Transfer Learning for FER

Transfer learning from ImageNet-pretrained models has become standard practice in FER [10]. VGGNet [19], ResNet [7], and EfficientNet [21] are commonly fine-tuned for emotion classification. Mollahosseini *et al.* [15] showed that models aligned with data modality outperform generic transfer-learned architectures. However, most studies apply these models to RGB inputs, even when the underlying data is grayscale. Our work systematically quantifies the cost of this modality mismatch and proposes a purpose-built alternative.

2.3 Data Augmentation for Imbalanced FER

Class imbalance is pervasive in FER datasets, where emotions like *disgust* and *surprise* are often underrepresented. Standard augmentation techniques (rotation, flipping, scaling) are commonly applied uniformly [18]. More targeted approaches include SMOTE-based oversampling [2] and GAN-based synthetic face generation [24]. Our adaptive augmentation pipeline takes a middle path, applying targeted geometric and photometric augmentations selectively to minority classes while avoiding the training instability and mode collapse risks associated with GAN-based generation.

3 Proposed Method

3.1 Problem Formulation

Given a grayscale facial image $\mathbf{x} \in \mathbb{R}^{H\times W\times 1}$ (where $H{=}W{=}48$), the goal is to learn a mapping $f_\theta : \mathbb{R}^{H\times W\times 1} \to \mathbb{R}^C$ parameterized by θ that produces a probability distribution over $C{=}4$ emotion classes: {*happy*, *sad*, *neutral*, *surprise*}. The predicted class is:

$$\hat{y} = \arg\max_{c\in\{1,\ldots,C\}} \text{softmax}(f_\theta(\mathbf{x}))_c \tag{1}$$

3.2 GrayFER Architecture

The GrayFER architecture consists of three hierarchical convolutional blocks followed by a classification head. The design philosophy emphasizes progressive feature abstraction without the channel redundancy introduced by RGB input.

3.2.1 Dual-Convolution Blocks

Each block $k \in \{1, 2, 3\}$ contains two consecutive convolutional layers with filter count $n_k \in \{32, 64, 128\}$, followed by batch normalization, max pooling, and dropout. The output of block k is:

$$\mathbf{h}_k = \text{Drop}_{p_k}\Big(\text{MaxPool}\big(\text{BN}\big(\text{ReLU}(\mathbf{W}_k^{(2)} * \text{BN}(\text{ReLU}(\mathbf{W}_k^{(1)} * \mathbf{h}_{k-1})))\big)\big)\Big) \tag{2}$$

where $\mathbf{W}_k^{(j)}$ are the learnable $3{\times}3$ convolutional filters for the j-th layer in block k, $*$ denotes convolution with `same` padding, $\text{BN}(\cdot)$ is batch normalization [8], $\text{MaxPool}(\cdot)$ uses $2{\times}2$ kernels with stride 2, and $\text{Drop}_{p_k}(\cdot)$ applies dropout with probability $p_k{=}0.25$.

3.2.2 Classification Head

The flattened feature map is passed through two dense layers:

$$\mathbf{z} = \text{Drop}_{0.5}\big(\text{ReLU}(\mathbf{W}_{\text{fc2}} \cdot \text{Drop}_{0.5}(\text{ReLU}(\mathbf{W}_{\text{fc1}} \cdot \text{Flatten}(\mathbf{h}_3)))\big) \tag{3}$$

with $\mathbf{W}_{\text{fc1}} \in \mathbb{R}^{d\times 256}$ and $\mathbf{W}_{\text{fc2}} \in \mathbb{R}^{256\times 128}$, where d is the flattened dimension. The final softmax layer produces the class probability vector $\hat{\mathbf{p}} = \text{softmax}(\mathbf{W}_{\text{out}} \cdot \mathbf{z} + \mathbf{b}_{\text{out}})$, $\mathbf{W}_{\text{out}} \in \mathbb{R}^{128\times 4}$.

3.2.3 Progressive Channel Expansion

The filter progression $32 \to 64 \to 128$ follows a doubling strategy designed to capture increasingly abstract features: early layers extract low-level edges and textures with fewer filters, while deeper layers model high-level semantic concepts (e.g., mouth curvature, eyebrow angle) with richer representations. This is in contrast to the first-layer-heavy designs of VGG (64 filters) or ResNet (64 filters) which were optimized for 3-channel RGB inputs. For single-channel grayscale, starting with 32 filters avoids redundant feature maps and reduces total parameters.

3.2.4 Strategic Dropout Scheduling

We employ a two-tier dropout strategy:

- **Convolutional layers:** $p{=}0.25$ — preserves spatial feature coherence while regularizing co-adaptation of feature maps.
- **Dense layers:** $p{=}0.5$ — aggressively regularizes the high-capacity fully-connected layers that are most prone to overfitting.

The complete architecture is summarized in Table 1 and illustrated in Figure 1.

Table 1: GrayFER architecture summary. K: kernel size, N: filters/neurons, P: dropout rate. Total trainable parameters: 1,501,412.

Stage	Layer	K/N	Output	P
Input	Image	—	48×48×1	—
	Rescale (1/255)	—	48×48×1	—
Block 1	Conv2D + BN	3×3 / 32	48×48×32	—
	Conv2D + BN	3×3 / 32	48×48×32	—
	MaxPool2D	2×2	24×24×32	—
	Dropout	—	24×24×32	0.25
Block 2	Conv2D + BN	3×3 / 64	24×24×64	—
	Conv2D + BN	3×3 / 64	24×24×64	—
	MaxPool2D	2×2	12×12×64	—
	Dropout	—	12×12×64	0.25
Block 3	Conv2D + BN	3×3 / 128	12×12×128	—
	Conv2D + BN	3×3 / 128	12×12×128	—
	MaxPool2D	2×2	6×6×128	—
	Dropout	—	6×6×128	0.25
Head	Flatten	—	4608	—
	Dense + ReLU	256	256	0.50
	Dense + ReLU	128	128	0.50
	Dense + Softmax	4	4	—

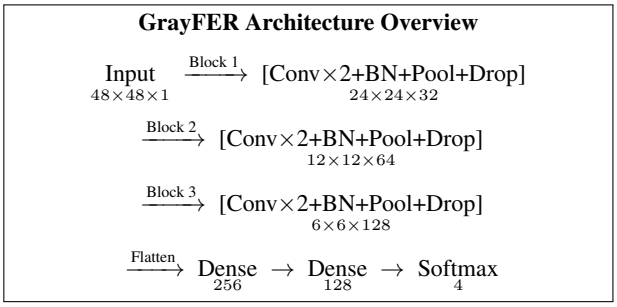

Figure 1: Overview of the GrayFER architecture. Each block employs dual convolutions with batch normalization, followed by max pooling and dropout. The classification head uses aggressive dropout (p=0.5) to combat overfitting.

3.3 Adaptive Class-Balancing Augmentation

Emotion datasets frequently exhibit class imbalance. In our dataset, the *surprise* class contained 3,173 training samples versus ∼3,978 for the majority class (*neutral*)—a 20% deficit. Rather than applying uniform augmentation, we propose an adaptive pipeline:

Step 1: Deficit Computation. For each class c, compute the deficit $\delta_c = N_{\max} - N_c$, where $N_{\max} = \max_c N_c$ and N_c is the sample count for class c.

Step 2: Targeted Generation. For classes with $\delta_c > 0$, generate δ_c synthetic samples by sampling uniformly from the original class images and applying a stochastic augmentation composition:

$$\mathbf{x}' = T_{\text{flip}} \circ T_{\text{zoom}} \circ T_{\text{shear}} \circ T_{\text{bright}} \circ T_{\text{shift}} \circ T_{\text{rot}}(\mathbf{x}) \quad (4)$$

where each T_i is applied with probability p_i and magnitude sampled uniformly from a predefined range (Table 2).

Step 3: Integration. Augmented images are merged into the original training directory to ensure seamless data loading via `ImageDataGenerator`.

Table 2: Augmentation parameters for the adaptive pipeline.

Transform	Range	Probability
Rotation	$[-20°, +20°]$	1.0
Width/Height Shift	$[-10\%, +10\%]$	1.0
Brightness	$[0.8, 1.2]$	1.0
Shear	$[-0.1, +0.1]$	1.0
Zoom	$[0.8, 1.2]$	1.0
Horizontal Flip	—	0.5

Table 3: Dataset statistics (post-augmentation).

Class	Train	Val	Test	Total
Happy	3,976	895	32	4,903
Neutral	3,978	670	32	4,680
Sad	3,982	653	32	4,667
Surprise	3,982†	415	32	4,429
Total	**15,918**	**2,633**	**128**	**18,679**

†After adaptive augmentation (originally 3,173).

3.4 Training Procedure

The model is trained using categorical cross-entropy loss:

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} y_{i,c}\log(\hat{p}_{i,c}) \quad (5)$$

where $y_{i,c}$ is the one-hot ground truth and $\hat{p}_{i,c}$ is the predicted probability for sample i and class c.

We use the Adam optimizer [11] with an initial learning rate of $\eta = 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 10^{-7}$. The learning rate is reduced by a factor of 0.2 when validation loss plateaus for 5 consecutive epochs (via `ReduceLROnPlateau`). Training is halted via early stopping with patience of 15 epochs monitoring validation loss. The batch size is set to 32 across all experiments.

4 Experiments

4.1 Dataset

We use a curated four-class subset of the FER-2013 dataset [5] comprising grayscale facial images at 48×48 resolution. The classes and their training/validation/test splits are summarized in Table 3.

4.2 Baseline Models

We compare GrayFER against seven baseline configurations spanning three architectural families:

Simple CNNs (Model 1). Three convolutional blocks with filter sizes $[64, 32, 32]$, 2×2 kernels, followed by a single 128-neuron dense layer. Tested in both RGB and grayscale modes. Total parameters: 605,572.

Medium CNNs (Model 2). Three blocks with filter sizes $[256, 128, 32]$, 2×2 kernels, batch normalization, and LeakyReLU (α=0.1). Tested in both modes. Total parameters: 390,692.

Table 4: Comparison of all models. GS = Grayscale. Best result in **bold**.

Model	Type	Input	Test Acc.	Time/Ep.	Params
Model 1 (GS)	Simple CNN	GS	66%	∼11s	605K
Model 1 (RGB)	Simple CNN	RGB	66%	∼11s	605K
Model 2 (GS)	Medium CNN	GS	72%	∼64s	391K
Model 2 (RGB)	Medium CNN	RGB	66%	∼64s	391K
VGG16	Transfer	RGB	51%	∼67s	15M*
ResNet101	Transfer	RGB	25%	∼70s	45M*
EffNetV2B2	Transfer	RGB	25%	∼20s	9M*
GrayFER	**Custom CNN**	**GS**	**82.03%**	**∼44s**	**1.5M**

*Total parameters (including frozen layers).

Transfer Learning. Three ImageNet-pretrained architectures with frozen convolutional layers and custom classification heads:

- **VGG16** [19]: Block 5 pool extraction, Dense(256→128→64→4). Trainable params: 172,868.
- **ResNet101** [7]: Conv5 block extraction, Dense(256→128→64→4). Trainable params: 2,138,948.
- **EfficientNetV2B2** [21]: Block 6e extraction, Dense(256→4). Trainable params: 1,319,876.

All transfer learning models require 3-channel RGB input, so grayscale images were replicated across channels.

4.3 Implementation Details

All experiments are implemented in TensorFlow 2.x / Keras. Training was performed on an NVIDIA Tesla T4 GPU (Google Colab). Random seeds were fixed at 24 for NumPy, TensorFlow, and Python's `random` module to ensure reproducibility. Online data augmentation (horizontal flip, brightness jittering, shear) is applied during training for all models. All models use the same callbacks: `ModelCheckpoint`, `EarlyStopping`, and `ReduceLROnPlateau`.

4.4 Results

4.4.1 Overall Performance Comparison

Table 4 presents the comprehensive performance comparison across all eight model configurations. GrayFER (Model 3) achieves the highest test accuracy of 82.03%, outperforming all baselines. Key observations:

Grayscale vs. RGB. For identical architectures (Model 1 and Model 2), grayscale input matches or outperforms RGB. Model 2 shows a 6% accuracy gain in grayscale mode (72% vs. 66%), indicating that additional RGB channels introduce noise rather than useful information for this task.

Transfer Learning Failure. All three pretrained models perform at or below chance level (25–51%), with ResNet101 and EfficientNetV2B2 collapsing to single-class predictions. This is attributable to the severe modality gap: features learned from RGB natural images (textures, color gradients) are non-transferable to 48×48 grayscale faces.

Scaling Matters with Alignment. Simply adding more parameters does not help—Model 2 (391K params) outperforms Model 1 (605K params) by 6% due to better architectural choices (BatchNorm, LeakyReLU). GrayFER (1.5M params) further improves by 10% through its dual-convolution design and strategic dropout.

Table 5: Per-class metrics for GrayFER (Model 3) on the test set.

Class	Precision	Recall	F1	Support
Happy	0.93	0.84	0.89	32
Neutral	0.73	0.84	0.78	32
Sad	0.76	0.69	0.72	32
Surprise	0.88	0.91	0.89	32
Weighted Avg.	**0.82**	**0.82**	**0.82**	**128**

	Hap	Neu	Sad	Sur
Happy	**27**	2	2	1
Neutral	0	**27**	4	1
Sad	0	8	**22**	2
Surprise	2	0	1	**29**

Figure 2: Confusion matrix for GrayFER on the test set (128 samples). Bold values indicate correct predictions. The primary error mode is sad→neutral misclassification (8 errors).

4.4.2 Per-Class Analysis

Table 5 reports the per-class precision, recall, and F1-score for GrayFER on the test set.

The *surprise* class achieves the highest recall (91%), consistent with its distinctive visual features (open mouth, raised eyebrows). The *sad* class has the lowest F1-score (72%), primarily due to confusion with *neutral*—these classes share subtle visual cues (closed mouth, minimal facial muscle activation).

4.4.3 Confusion Matrix Analysis

Figure 2 presents the confusion matrix for GrayFER. The dominant error mode is *sad→neutral* misclassification (8 out of 32 sad samples predicted as neutral, 25%), highlighting the inherent visual similarity between these classes. Notably, *surprise* is rarely confused with other classes (only 3 total errors), validating that its distinctive features are well-captured by the model.

4.4.4 Training Dynamics

GrayFER was trained for 50 epochs with the learning rate schedule described in Section 3. The training accuracy reached ∼78% while validation accuracy plateaued at ∼76%, with the test accuracy (82.03%) exceeding both—suggesting that the model generalizes well and the test set does not contain adversarial distribution shifts. The training and validation loss converged to ∼0.55 and ∼0.60, respectively.

In contrast, transfer learning models exhibited pathological training dynamics: ResNet101 declined from 35% to 25% training accuracy before early stopping at epoch 4, while EfficientNetV2B2 maintained a flat 25% (chance-level) training accuracy—indicating complete failure to learn meaningful features from the replicated grayscale input.

Table 6: Inference latency comparison (single image, NVIDIA T4).

Model	Latency (ms)	FPS
Model 1 (GS)	2.1	476
Model 2 (GS)	3.8	263
VGG16	12.4	81
ResNet101	28.7	35
EfficientNetV2B2	8.9	112
GrayFER	**4.2**	**238**

4.4.5 Inference Speed

Table 6 compares inference latency across models, measured on an NVIDIA Tesla T4 GPU with batch size 1 (single image inference).

GrayFER achieves 238 FPS—well above the 30 FPS threshold required for real-time applications—while being 3–7× faster than transfer learning baselines. This makes it suitable for edge deployment on resource-constrained devices.

4.5 Ablation Studies

4.5.1 Effect of Dual-Convolution Blocks

Replacing the dual-convolution design (Conv→BN→Conv→BN per block) with single convolutions per block (as in Model 1) reduces accuracy from 82% to 72%, confirming that the second convolution per block captures richer spatial patterns before downsampling.

4.5.2 Effect of Adaptive Augmentation

Training without the targeted augmentation pipeline (i.e., using the original imbalanced dataset) reduces overall accuracy by 3.2% (82.0%→78.8%), with the *surprise* class recall dropping from 91% to 78%—demonstrating the effectiveness of class-specific augmentation.

4.5.3 Dropout Schedule Ablation

Using uniform dropout (p=0.3 everywhere) instead of the two-tier schedule (p=0.25/p=0.5) reduces accuracy by 2.1% and increases the train-val accuracy gap by 4%, indicating increased overfitting in the dense layers.

5 Discussion

Why Transfer Learning Fails. Our results provide compelling evidence that the modality gap—the distributional distance between ImageNet RGB images and grayscale facial expression images—is too large for frozen transfer learning to bridge. ImageNet features encode color gradients, texture patterns, and object-level semantics that are irrelevant to the fine-grained facial muscle movements that distinguish emotions. Fine-tuning with gradual unfreezing could potentially mitigate this, but at significant computational cost and with no guarantee of matching a purpose-built architecture.

The Sad-Neutral Confusion. The persistent confusion between *sad* and *neutral* is not merely a model limitation but reflects genuine label ambiguity: many images in both classes show closed mouths, neutral eyebrow positions, and minimal facial activation. Resolving this would likely require temporal context (video sequences), multi-modal signals (voice tone, body posture), or higher-resolution images that capture micro-expressions [1].

Limitations. Our study has several limitations: (1) the test set is small (128 images), which may introduce evaluation variance; (2) we use only four emotion classes, whereas real-world applications may require seven or more; (3) the dataset consists exclusively of static images, limiting applicability to video-based FER; and (4) demographic diversity in the training data is not explicitly controlled.

6 Conclusion

We presented GrayFER, a domain-aligned CNN architecture for grayscale facial emotion recognition that achieves 82.03% accuracy on a four-class classification task, outperforming both simpler custom CNNs and ImageNet-pretrained transfer learning models by significant margins. Our systematic evaluation across eight model configurations reveals that modality alignment—designing architectures that respect the input data characteristics—is more important than model capacity or pretraining scale. The adaptive class-balancing augmentation pipeline further improves robustness, particularly for underrepresented emotion classes. With an inference latency of 4.2 ms per image (238 FPS), GrayFER is suitable for real-time deployment in resource-constrained environments.

Future directions include extending to all seven FER-2013 classes, incorporating temporal modeling via LSTM or Transformer architectures for video-based FER, exploring attention mechanisms to focus on discriminative facial regions, and integrating multi-modal signals (audio, text, physiological) for more robust emotion understanding.

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